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Turbine and turbocharger

Abstract

A turbine includes: a turbine scroll flow path; a turbine blade wheel disposed on a radially inner side with respect to the turbine scroll flow path; and a blade included in the turbine blade wheel, the blade having a leading edge inclined to a side opposite to a rotation direction side of the turbine blade wheel as the leading edge extends from a hub side to a shroud side, the leading edge having an inclination angle greater than 0° and less than or equal to 45° with respect to an axial direction of the turbine blade wheel as viewed in the radial direction.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation application of International Application No. PCT/JP2022/006385, filed on Feb. 17, 2022, which claims priority to Japanese Patent Application No. 2021-044156, filed on Mar. 17, 2021, the entire contents of which are incorporated by reference herein.

BACKGROUND ART
Technical Field
(1) The present disclosure relates to a turbine and a turbocharger. The present application claims the benefit of priority based on Japanese Patent Application No. 2021-044156 filed on Mar. 17, 2021, the content of which is incorporated herein.
Related Art
(2) Some turbines included in a turbocharger and the like are of a type in which gas flows into a turbine blade wheel from a radially outer side. As such a type of turbine, for example, Patent Literature 1 discloses a radial turbine in which gas flows in a radial direction. Note that turbines into which gas flows in a direction inclined with respect to the radial direction are referred to as diagonal flow turbines.
CITATION LIST
Patent Literature
(3) Patent Literature 1: WO 2014/128898 A
SUMMARY
Technical Problem
(4) In turbines of a type in which gas flows into a turbine blade wheel from a radially outer side, at a gas inlet portion of the turbine blade wheel (namely, a portion of the turbine blade wheel into which the gas flows), depending on operation conditions, separation of the gas flow may occur, and a vortex may be generated. If separation of the gas flow occurs at the gas inlet portion in the turbine blade wheel, the efficiency of the turbine decreases.
(5) An object of the present disclosure is to provide a turbine and a turbocharger capable of improving efficiency of the turbine.
Solution to Problem
(6) In order to solve the above disadvantage, a turbine of the present disclosure includes: a turbine scroll flow path; a turbine blade wheel disposed on a radially inner side with respect to the turbine

scroll flow path; and a blade included in the turbine blade wheel, the blade having a leading edge inclined to a side opposite to a rotation direction side of the turbine blade wheel as the leading edge extends from a hub side to a shroud side, the leading edge having an inclination angle greater than 0° and less than or equal to 45° with respect to an axial direction of the turbine blade wheel as viewed in the radial direction.
(7) The inclination angle of the leading edge may be within a range of 10° to 30°.
(8) The leading edge may have a linear shape.
(9) In order to solve the above disadvantage, the turbocharger of the present disclosure includes the turbine described above.
Effects of Disclosure
(10) According to the present disclosure, efficiency of a turbine can be improved.

Description

BRIEF DESCRIPTION OF DRAWINGS
(1) FIG. **1** is a schematic cross-sectional view illustrating a turbocharger according to an embodiment of the present disclosure.
(2) FIG. **2** is a cross-sectional view taken along line A-A in FIG. **1**.
(3) FIG. **3** is a view extracted from a one-dot chain line part of FIG. **1**.
(4) FIG. **4** is a side view illustrating a turbine blade wheel according to an embodiment of the present disclosure.
(5) FIG. **5** is a graph illustrating the relationship between the inclination angle of a leading edge and the difference in efficiency of a turbine.
DESCRIPTION OF EMBODIMENTS
(6) Embodiments of the present disclosure will be described below by referring to the accompanying drawings. Dimensions, materials, other specific numerical values, and the like illustrated in the embodiments are merely an example for facilitating understanding, and the present disclosure is not limited thereto unless otherwise specified. Note that, in the present specification and the drawings, components having substantially the same function and structure are denoted by the same symbol, and redundant explanations are omitted. Illustration of components not directly related to the present disclosure is omitted.
(7) FIG. **1** is a schematic cross-sectional view of a turbocharger TC. Hereinafter, description is given on the premise that the direction of an arrow L illustrated in FIG. **1** is the left side of the turbocharger TC. Description is given on the premise that the direction of an arrow R illustrated in FIG. **1** is the right side of the turbocharger TC. As illustrated in FIG. **1**, the turbocharger TC includes a turbocharger main body **1**. The turbocharger main body **1** includes a bearing housing **3**, a turbine housing **5**, and a compressor housing **7**. The turbine housing **5** is connected to the left side of the bearing housing **3** by a fastening mechanism **9**. The compressor housing **7** is connected to the right side of the bearing housing **3** by a fastening bolt **11**. The turbocharger TC includes a turbine T and a centrifugal compressor C. The turbine T includes the bearing housing **3** and the turbine housing **5**. The centrifugal compressor C includes the bearing housing **3** and the compressor housing **7**.
(8) A protrusion **3***a* is formed on an outer curved surface of the bearing housing **3**. The protrusion **3***a* is formed on a side closer to the turbine housing **5**. The protrusion **3***a* protrudes in a radial direction of the bearing housing **3**. A protrusion **5***a* is formed on an outer curved surface of the turbine housing **5**. The protrusion **5***a* is formed on a side closer to the bearing housing **3**. The protrusion **5***a* protrudes in a radial direction of the turbine housing **5**. The bearing housing **3** and the turbine housing **5** are band-fastened to each other by the fastening mechanism **9**. The fastening mechanism **9** is, for example, a G coupling. The fastening mechanism **9** clamps the protrusions **3***a*

and **5***a*.
(9) A bearing hole **3***b* is formed in the bearing housing **3**. The bearing hole **3***b* penetrates through the turbocharger TC in the left-right direction. A bearing is disposed in the bearing hole **3***b*. A shaft **13** is inserted through the bearing. The bearing pivotally supports the shaft **13** in a freely rotatable manner. The bearing is a slide bearing. However, the bearing is not limited thereto and may be a rolling bearing. At a left end of the shaft **13**, a turbine blade wheel **15** is provided. The turbine blade wheel **15** is housed in the turbine housing **5** in a freely rotatable manner. At a right end of the shaft **13**, a compressor impeller **17** is provided. The compressor impeller **17** is accommodated in the compressor housing **7** in a freely rotatable manner.
(10) An intake port **19** is formed in the compressor housing **7**. The intake port **19** opens to the right side of the turbocharger TC. The intake port **19** is connected to an air cleaner (not illustrated). Facing surfaces of the bearing housing **3** and the compressor housing **7** constitute a diffuser flow path **21**. The diffuser flow path **21** pressurizes the air. The diffuser flow path **21** is formed in an annular shape. The diffuser flow path **21** communicates with the intake port **19** via the compressor impeller **17** on the radially inner side.
(11) A compressor scroll flow path **23** is formed in the compressor housing **7**. The compressor scroll flow path **23** is formed in an annular shape. The compressor scroll flow path **23** is positioned on an outer side in the radial direction of the shaft **13** with respect to the diffuser flow path **21**. The compressor scroll flow path **23** communicates with an intake port of an engine (not illustrated) and the diffuser flow path **21**. When the compressor impeller **17** rotates, the air is sucked from the intake port **19** into the compressor housing **7**. The sucked air is pressurized and accelerated in the process of flowing between blades of the compressor impeller **17**. The pressurized and accelerated air is further pressurized by the diffuser flow path **21** and the compressor scroll flow path **23**. The pressurized air is guided to the intake port of the engine.
(12) In the turbine housing **5**, a discharge flow path **25**, an accommodation unit **27**, and an exhaust flow path **29** are formed. The discharge flow path **25** opens to the left side of the turbocharger TC. The discharge flow path **25** is connected to an exhaust gas purification device (not illustrated). The discharge flow path **25** communicates with the accommodation unit **27**. The discharge flow path **25** is continuous with the accommodation unit **27** in the rotation axis direction of the turbine blade wheel **15**. The accommodation unit **27** accommodates the turbine blade wheel **15**. The exhaust flow path **29** is formed on a radially outer side with respect to the turbine blade wheel **15**. The exhaust flow path **29** is formed in an annular shape. The exhaust flow path **29** includes a turbine scroll flow path **29***a*. The turbine scroll flow path **29***a* communicates with the accommodation unit **27**. That is, the turbine blade wheel **15** is disposed on a radially inner side with respect to the turbine scroll flow path **29***a*.
(13) The exhaust flow path **29** communicates with an exhaust manifold of the engine (not illustrated). The exhaust gas discharged from the exhaust manifold of the engine (not illustrated) is guided to the discharge flow path **25** via the exhaust flow path **29** and the accommodation unit **27**. The exhaust gas guided to the discharge flow path **25** rotates the turbine blade wheel **15** in the process of flowing therethrough.
(14) The turning force of the turbine blade wheel **15** is transmitted to the compressor impeller **17** via the shaft **13**. When the compressor impeller **17** rotates, the air is pressurized as described above. In this manner, the air is guided to the intake port of the engine.
(15) FIG. **2** is a cross-sectional view taken along line A-A in FIG. **1**. In FIG. **2**, as for the turbine blade wheel **15**, only the outer circumference is illustrated as a circle. As illustrated in FIG. **2**, the exhaust flow path **29** is formed on the radially outer side of the accommodation unit **27** (namely, the radially outer side of the turbine blade wheel **15**). The exhaust flow path **29** includes the turbine scroll flow path **29***a*, a communicating portion **29***b*, an exhaust introduction port **29***c*, and an exhaust introduction path **29***d*.
(16) The communicating portion **29***b* is formed in an annular shape over the entire circumference

of the accommodation unit **27**. The turbine scroll flow path **29***a* is positioned on a radially outer side of the turbine blade wheel **15** with respect to the communicating portion **29***b*. The turbine scroll flow path **29***a* is annularly formed over the entire circumference of the communicating portion **29***b* (namely, the entire circumference of the accommodation unit **27**). The communicating portion **29***b* communicates the accommodation unit **27** with the turbine scroll flow path **29***a*. A tongue portion **31** is formed in the turbine housing **5**. The tongue portion **31** is provided at an end on the downstream side of the turbine scroll flow path **29***a* and partitions into a portion on the downstream side and a portion on the upstream side of the turbine scroll flow path **29***a*.
(17) The exhaust introduction port **29***c* opens to the outside of the turbine housing **5**. Exhaust gas discharged from the exhaust manifold of the engine (not illustrated) is introduced to the exhaust introduction port **29***c*. The exhaust introduction path **29***d* is formed between the exhaust introduction port **29***c* and the turbine scroll flow path **29***a*. The exhaust introduction path **29***d* connects the exhaust introduction port **29***c* and the turbine scroll flow path **29***a*. The exhaust introduction path **29***d* is formed, for example, in a linear shape. The exhaust introduction path **29***d* guides the exhaust gas introduced from the exhaust introduction port **29***c* to the turbine scroll flow path **29***a*. The turbine scroll flow path **29***a* guides the exhaust gas introduced from the exhaust introduction path **29***d* to the accommodation unit **27** via the communicating portion **29***b*.
(18) A bypass flow path **33** is formed in the turbine housing **5**. An inlet end OP of the bypass flow path **33** opens to the exhaust flow path **29** (specifically, the exhaust introduction path **29***d*). An outlet end of the bypass flow path **33** opens to the discharge flow path **25** (see FIG. **1**). The bypass flow path **33** communicates (connects) the exhaust introduction path **29***d* and the discharge flow path **25**. A wastegate port WP (see FIG. **1**) is formed at the outlet end of the bypass flow path **33**. A wastegate valve WV (see FIG. **1**) capable of opening and closing the wastegate port WP is disposed at the outlet end of the bypass flow path **33**. The wastegate valve WV is arranged inside the discharge flow path **25**. When the wastegate valve WV opens the wastegate port WP, the bypass flow path **33** allows a part of the exhaust gas flowing through the exhaust introduction path **29***d* to bypass the accommodation unit **27** (namely, the turbine blade wheel **15**) and to flow into the discharge flow path **25**.
(19) In the turbine T, with the opening and closing operation of the wastegate port WP controlled, the flow rate of the exhaust gas flowing into the turbine blade wheel **15** is adjusted. As described above, the turbine T is a variable capacity turbine. The bypass flow path **33** and the wastegate valve WV correspond to a flow rate adjusting mechanism for adjusting the flow rate of the exhaust gas flowing into the turbine blade wheel **15**. However, the flow rate adjusting mechanism is not limited to the above example as described later.
(20) FIG. **3** is an extracted diagram of an alternate long and short dash line portion of FIG. **1**. As illustrated in FIG. **3**, the turbine blade wheel **15** includes a hub **15***a* and a plurality of blades **15***b*. Hereinafter, the axial direction, the circumferential direction, and the radial direction of the turbine blade wheel **15** are also simply referred to as the axial direction, the circumferential direction, and the radial direction. The hub **15***a* is connected with the left end of the shaft **13** (see FIG. **1**). The outer diameter of the hub **15***a* decreases as it is closer to the left side of the turbocharger TC. A plurality of blades **15***b* is provided on the outer curved surface of the hub **15***a*. The plurality of blades **15***b* are provided at intervals in the circumferential direction. A blade **15***b* is formed to extend radially outward from the outer curved surface of the hub **15***a*. The outer edge of the blade **15***b* includes a leading edge LE and a trailing edge TE.
(21) The leading edge LE is an upstream edge of the blade **15***b* in the flow direction of the exhaust gas. The leading edge LE is an edge of the blade **15***b* on the turbine scroll flow path **29***a* side. Exhaust gas flows toward the leading edge LE from the turbine scroll flow path **29***a*. That is, the portion of the turbine blade wheel **15** where leading edges LE are arrange corresponds to the inlet portion (namely, a portion of the turbine blade wheel **15** into which the exhaust gas flows) of the exhaust gas in the turbine blade wheel **15**. The leading edge LE is formed on the right end side of

the blade **15***b*. The leading edge LE extends in the axial direction of the turbine blade wheel **15** when viewed in the circumferential direction. In the example of FIG. **3**, the leading edge LE is inclined radially outward as it extends in the axial direction. However, the leading edge LE may be parallel to the axial direction when viewed in the circumferential direction.
(22) The right end of the leading edge LE is a hub-side end P1 (namely, an end on the hub **15***a* side). The left end of the leading edge LE is a shroud-side end P2 (namely, an end on a shroud **27***a* side which is a portion of the turbine housing **5**, the portion forming the accommodation unit **27**). The leading edge LE extends from the hub-side end P1 to the shroud-side end P2.
(23) The trailing edge TE is a downstream edge of the blade **15***b* in the flow direction of the exhaust gas. The trailing edge TE is an edge of the blade **15***b* on the discharge flow path **25** side. The exhaust gas flows out from the trailing edge TE toward the discharge flow path **25**. The trailing edge TE is formed on the left end side of the blade **15***b*. The trailing edge TE extends in the radial direction when viewed in the circumferential direction. Specifically, the trailing edge TE extends in the radial direction while being twisted in the circumferential direction.
(24) A portion of the outer peripheral edge of the blade **15***b* between the leading edge LE and the trailing edge TE extends along the shroud **27***a* of the turbine housing **5**.
(25) As indicated by an arrow FD in FIG. **3**, the exhaust gas flows into the turbine blade wheel **15** from the radially outer side. As described above, the turbine T is a radial turbine.
(26) Note that the turbine T may be a diagonal flow turbine into which the exhaust gas flows from the radially outer side in a direction inclined with respect to the radial direction.
(27) Here, in the turbine T of the type in which gas flows into the turbine blade wheel **15** from the radially outer side, at the inlet portion of the exhaust gas in the turbine blade wheel **15**, depending on operation conditions, separation of the gas flow may occur, and a vortex may be generated. In the turbine T which is a variable capacity turbine, operating conditions (specifically, for example, the flow rate of the exhaust gas flowing into the turbine blade wheel **15**) vary over a wide range. Therefore, in the turbine T which is a variable capacity turbine, separation of the gas flow at the inlet portion of the exhaust gas in the turbine blade wheel is particularly likely to occur. Such separation of the gas flow causes a decrease in the efficiency of the turbine
(28) T. In the turbine T which is a variable capacity turbine, there is a particularly high need to improve the efficiency of the turbine T.
(29) Therefore, in the turbine T according to the present embodiment, the shapes of the blades **15***b* of the turbine blade wheel **15** are devised in order to improve the efficiency of the turbine T. Hereinafter, the shape of a blade **15***b* of the turbine blade wheel **15** will be described in detail with reference to FIGS. **4** and **5**.
(30) FIG. **4** is a side view illustrating the turbine blade wheel **15** according to the embodiment. A leading edge LE of a blade **15***b* of the turbine blade wheel **15** is inclined to a side opposite to a rotation direction RD side of the turbine blade wheel **15** as it extends from the hub **15***a* side to the shroud **27***a* side. That is, in each of the blades **15***b*, the circumferential position of a hub-side end P1 of a leading edge LE is advanced in the rotation direction RD with respect to the circumferential position of a shroud-side end P2 of the leading edge LE. In the example of FIG. **4**, the rotation direction RD is a counterclockwise direction when the turbine blade wheel **15** is viewed from the left side (namely, from the above in FIG. **4**) of the turbocharger TC.
(31) A leading edge LE has a linear shape. Specifically, a leading edge LE extends along a straight line connecting a hub-side end P1 and a shroud-side end P2. However, the shape of the leading edge LE is not limited to the linear shape. For example, a part of the leading edge LE may be curved or bent.
(32) The inventor has found that the efficiency of the turbine T varies depending on the inclination angle α1 of the leading edge LE with respect to an axial direction AD of the turbine blade wheel **15** when viewed in the radial direction by performing flow analysis simulation. In the flow analysis simulation, the state of the gas flow in the turbine blade wheel **15** when the inclination angle α1 is

varied in variously manners (for example, the direction, the velocity, the entropy, and others) and the efficiency of the turbine T were calculated. In particular, it became clear from the flow analysis simulation that the efficiency of the turbine T is improved in a case where the leading edge LE is inclined to the side opposite to the rotation direction RD side of the turbine blade wheel **15** as the leading edge LE extends from the hub **15***a* side to the shroud **27***a* side and the inclination angle α1 is set within a specific range. As a result, in the turbine T according to the present embodiment, the inclination angle α1 of the leading edge LE is larger than 0° and equal to or smaller than 45°. As a result, the efficiency of the turbine T is improved.
(33) FIG. **5** is a graph illustrating the relationship between the inclination angle α1 [deg] of the leading edge LE and the difference in efficiency ΔE [%] of the turbine T. FIG. **5** is a graph obtained by the flow analysis simulation. The difference in efficiency ΔE of the turbine T is a change amount of the efficiency of the turbine T at each inclination angle α1 with respect to the efficiency of the turbine T of a case where the inclination angle α1 is 0°. That is, the difference in efficiency ΔE is obtained by subtracting the efficiency of the turbine T of the case where the inclination angle α1 is 0° from the efficiency of the turbine T at each inclination angle α1. The efficiency of the turbine T is the ratio of energy generated by the turbine T to energy input to the turbine T.
(34) According to the graph illustrated in FIG. **5**, it can be seen that the efficiency of the turbine T is higher in a case where the inclination angle α1 of the leading edge LE is larger than 0° and less than or equal to 45° as compared to a case where the inclination angle α1 is less than or equal to 0° or a case where the inclination angle α1 is larger than 45°. Note that the case where the inclination angle α1 is less than or equal to 0° is a case where the leading edge LE is parallel to the axial direction AD or a case where the leading edge LE is inclined toward the rotation direction RD side of the turbine blade wheel **15** as it extends from the hub **15***a* side to the shroud **27***a* side.
(35) In the flow analysis simulation, it was observed that in the case where the inclination angle α1 of the leading edge LE is larger than 0° and less than or equal to 45°, separation of the gas flow and generation of a vortex caused by the separation are suppressed in the portion where the leading edge LE is disposed in the turbine blade wheel **15** (namely, the inlet portion of the exhaust gas in the turbine blade wheel **15**). That is, it is conceivable that, in the case where the inclination angle α1 of the leading edge LE is larger than 0° and less than or equal to 45°, the separation of the gas flow is suppressed, and thus, as a result, the efficiency of the turbine T is improved.
(36) According to the flow analysis simulation, it can be seen that in a case where the leading edge LE is inclined to the side opposite to the rotation direction RD side of the turbine blade wheel **15** as it extends from the hub **15***a* side to the shroud **27***a* side, the impact when the exhaust gas flowing in from the leading edge LE collides with the blade **15***b* is mitigated, and generation of a vortex of the flow of the exhaust gas is suppressed. Meanwhile, it can be seen that in a case where the inclination angle α1 of the leading edge LE is excessively large, the flow of the exhaust gas is less likely to follow along the blade **15***b* after the exhaust gas flowing in from the leading edge LE collides with the blade **15***b*, and separation of the gas flow is more likely to occur.
(37) In the graph illustrated in FIG. **5**, in a case where the inclination angle α1 is around 20°, the difference in efficiency ΔE of the turbine T has the maximum value. As described above, according to the graph illustrated in FIG. **5**, it can be seen that the efficiency of the turbine T is particularly improved in a case where the inclination angle α1 of the leading edge LE is within a range of 10° to 30°.
(38) In the flow analysis simulation, it was observed that in the case where the inclination angle α1 of the leading edge LE is within the range of 10° to 30°, separation of the gas flow and generation of a vortex caused by the separation are effectively suppressed in the portion where the leading edge LE is disposed in the turbine blade wheel **15** (namely, the inlet portion of the exhaust gas in the turbine blade wheel **15**). That is, it is conceivable that, in the case where the inclination angle α1 of the leading edge LE is within the range of 10° to 30°, the separation of the gas flow is efficiency suppressed, and thus the efficiency of the turbine T is effectively improved.

(39) As described above, in the turbine T, a leading edge LE has a linear shape. This makes it possible to appropriately implement improvement of the efficiency of the turbine T by optimizing the inclination angle α1 on the basis of the knowledge obtained from the flow analysis simulation.
(40) Although the embodiments of the present disclosure has been described with reference to the accompanying drawings, it is naturally understood that the present disclosure is not limited to the above embodiments. It is clear that those skilled in the art can conceive various modifications or variations within the scope described in the claims, and it is understood that they are naturally also within the technical scope of the present disclosure.
(41) In the above description, the example in which the turbine T is of a single scroll type (a type in which the number of turbine scroll flow paths **29***a* is one) has been described, however, the type of the turbine T is not limited to the above example. For example, the turbine T may be of a double scroll type (a type in which two turbine scroll flow paths **29***a* are connected with the accommodation unit **27** at different circumferential positions) or of a twin scroll type (a type in which two turbine scroll flow paths **29***a* are arranged side by side in the axial direction).
(42) The example in which the bypass flow path **33** and the wastegate valve WV are used as the flow rate adjusting mechanism for adjusting the flow rate of the exhaust gas flowing into the turbine blade wheel **15** has been described above. However, the flow rate adjusting mechanism is not limited to the above example. For example, as the flow rate adjusting mechanism, a mechanism including a plurality of variable nozzle blades capable of adjusting the cross-sectional area of a flow path on the upstream side with respect to the turbine blade wheel **15** may be used. The plurality of variable nozzle blades is provided on the radially outer side with respect to the turbine blade wheel **15**. The plurality of variable nozzle blades is provided at intervals in the circumferential direction of the turbine blade wheel **15**. As the variable nozzle blades rotate, the cross-sectional area of the flow path on the upstream side of the turbine blade wheel **15** varies depending on the rotation angle of the variable nozzle blades. As a result, the flow rate of the exhaust gas flowing into the turbine blade wheel is adjusted. Note that the turbine T may not be provided with the flow rate adjusting mechanism. Both the flow rate adjusting mechanism including the wastegate valve WV and the flow rate adjusting mechanism including the variable nozzle blades may be provided to the turbine T.
(43) The example in which the turbine T is included in the turbocharger TC has been described above. However, the turbine T may be included in a device other than the turbocharger TC.

Claims

1. A turbine comprising: a turbine scroll flow path; a turbine blade wheel disposed on a radially inner side with respect to the turbine scroll flow path; and a blade comprised in the turbine blade wheel, the blade including a leading edge inclined to a side opposite to a rotation direction side of the turbine blade wheel as the leading edge extends from a hub side to a shroud side, the leading edge having an inclination angle with respect to an axial direction of the turbine blade wheel as viewed in a radial direction within a range of 10° to 30°, the leading edge has a linear shape, and the blade includes a trailing edge having a linear shape, wherein the inclination angle is constant from the hub side to the shroud side.
2. A turbocharger comprising the turbine according to claim 1.